

Lecture 7 Pre-CTMC

From short-time transition rules to exponential holding times

Short-time law

$$P(\Delta t) = I + Q\Delta t + o(\Delta t)$$

Semigroup and matrix exponential

Holding time: $\tau_i \sim \text{Exp}(\Lambda_i)$

A reorganized version following the board derivation: local transition probabilities, generator, semigroup, and first jump time.

Roadmap

Goal

Starting from the local short-time transition probabilities of a continuous-time Markov chain, we will first identify the generator entries from limits, then derive the matrix evolution law, and finally study the first jump time from a fixed state.

Part I: Global transition law

1. short-time probabilities with rates λ_{ij}
2. define q_{ij} and q_{ii} from limits
3. only then write the matrix forms for $P(t)$ and Q

Part II: When to jump

1. define the first jump time τ_i
2. derive its survival function
3. obtain density and mean waiting time

Short-time transition assumption

Let the state space be a countable set $I = \{0, 1, 2, \dots\}$. For a CTMC, the future depends only on the current state.

Local transition rule at state i

For $j \neq i$, in a very short interval Δt ,

$$\mathbb{P}(X_{t+\Delta t} = j \mid X_t = i) = \lambda_{ij}\Delta t + o(\Delta t).$$

Hence the probability of staying at i is

$$\mathbb{P}(X_{t+\Delta t} = i \mid X_t = i) = 1 - \sum_{j \neq i} \lambda_{ij}\Delta t + o(\Delta t).$$

Notation

Define the total leaving rate

$$\Lambda_i := \sum_{j \neq i} \lambda_{ij}.$$

From λ_{ij} to q_{ij}

Off-diagonal terms

For $j \neq i$,

$$\frac{\mathbb{P}(X_{t+\Delta t} = j \mid X_t = i) - \mathbb{P}(X_t = j \mid X_t = i)}{\Delta t} = \frac{\lambda_{ij}\Delta t + o(\Delta t) - 0}{\Delta t}.$$

Taking $\Delta t \rightarrow 0$ gives

$$q_{ij} = \lambda_{ij}, \quad j \neq i.$$

Diagonal term

$$q_{ii} = \lim_{\Delta t \rightarrow 0} \frac{\mathbb{P}(X_{t+\Delta t} = i \mid X_t = i) - 1}{\Delta t}.$$

Using

$$\mathbb{P}(X_{t+\Delta t} = i \mid X_t = i) = 1 - \Lambda_i \Delta t + o(\Delta t),$$

we obtain

$$q_{ii} = -\Lambda_i = -\sum_{j \neq i} \lambda_{ij}.$$

Only now introduce the generator matrix

After defining the entries one by one, collect them into the generator

$$Q = (q_{ij}).$$

The entrywise short-time formulas become

$$\mathbb{P}(X_{t+\Delta t} = j \mid X_t = i) = q_{ij}\Delta t + o(\Delta t), \quad j \neq i,$$

and

$$\mathbb{P}(X_{t+\Delta t} = i \mid X_t = i) = 1 + q_{ii}\Delta t + o(\Delta t).$$

Generator structure

$$q_{ij} \geq 0 \ (j \neq i), \quad q_{ii} = - \sum_{j \neq i} q_{ij}.$$

Why this is intuitive

Off-diagonal entries are jump rates to other states. The diagonal entry is exactly what is needed so that each row still sums to 1 to first order.

The infinitesimal form of the transition matrix

Now define the finite-time transition probabilities

$$P_{ij}(t) := \mathbb{P}(X_t = j \mid X_0 = i).$$

Combining the diagonal and off-diagonal cases gives

$$P_{ij}(\Delta t) = \delta_{ij} + q_{ij}\Delta t + o(\Delta t).$$

In matrix form,

$$P(\Delta t) = I + Q\Delta t + o(\Delta t),$$

where $Q = (q_{ij})$ is the generator matrix.

Interpretation

The matrix form is just a compact summary of the entrywise limits:

- off-diagonal entries record where the chain can jump,
- diagonal entries record how quickly the chain leaves the current state.

Matrix differential equation

By the Markov property,

$$P(t + \Delta t) = P(t) P(t + \Delta t \mid t).$$

Time-homogeneity implies

$$P(t + \Delta t \mid t) = P(\Delta t).$$

Hence

$$P(t + \Delta t) = P(t)P(\Delta t) = P(t)(I + Q\Delta t + o(\Delta t)).$$

Rearranging,

$$\frac{P(t + \Delta t) - P(t)}{\Delta t} = P(t)Q + o(1).$$

Letting $\Delta t \rightarrow 0$,

$$P'(t) = P(t)Q.$$

Kolmogorov equation in matrix form

With initial condition $P(0) = I$,

$$P'(t) = P(t)Q, \quad P(0) = I.$$

Why the solution is $P(t) = e^{tQ}$

Consider the matrix exponential

$$e^{tQ} := \sum_{n=0}^{\infty} \frac{(tQ)^n}{n!}.$$

Differentiate term by term:

$$\frac{d}{dt} e^{tQ} = Q \sum_{n=0}^{\infty} \frac{t^n Q^n}{n!} = e^{tQ} Q.$$

Also,

$$e^{0Q} = I.$$

Therefore e^{tQ} solves the same differential equation and initial condition:

$$P'(t) = P(t)Q, \quad P(0) = I.$$

Conclusion

$$P(t) = e^{tQ}.$$

This is the global transition law obtained from the local generator Q .

An equivalent discrete-time approximation

From

$$P(\Delta t) = I + Q\Delta t + o(\Delta t),$$

one can iterate over many tiny steps:

$$P(t) \approx \left(I + Q\frac{t}{n}\right)^n.$$

Letting $n \rightarrow \infty$ gives

$$P(t) = \lim_{n \rightarrow \infty} \left(I + Q\frac{t}{n}\right)^n = e^{tQ}.$$

Meaning

This matches the intuition from the board derivation: many infinitesimal updates combine into the matrix exponential.

Two important consequences of $P(t) = e^{tQ}$

1. Do not read e^{tQ} entrywise

In general,

$$P_{ii}(t) \neq e^{tq_{ii}}.$$

The expression e^{tQ} is a **matrix exponential**, not an entrywise exponential.

2. Probability is conserved

Because each row of Q sums to zero,

$$Q\mathbf{1} = \mathbf{0}.$$

Hence

$$P(t)\mathbf{1} = e^{tQ}\mathbf{1} = \mathbf{1}.$$

So every row of $P(t)$ sums to 1, exactly as a transition matrix should.

Now ask a different question: when to jump?

Fix a state i and assume the chain starts there:

$$X_0 = i.$$

Define the first jump time from state i by

$$\tau_i := \inf\{t > 0 : X_t \neq i\}.$$

Important distinction

$$\mathbb{P}(\tau_i > t \mid X_0 = i)$$

means the chain has stayed at i during the entire interval $[0, t]$.

This is stronger than merely saying $X_t = i$, because the chain could leave and return.

Survival function of the first jump time

Let

$$S_i(t) := \mathbb{P}(\tau_i > t \mid X_0 = i).$$

To survive up to time $t + \Delta t$, the chain must:

1. survive up to time t ;
2. then stay at i during $[t, t + \Delta t]$.

Hence

$$S_i(t + \Delta t) = S_i(t)(1 - \Lambda_i \Delta t + o(\Delta t)).$$

Therefore

$$\frac{S_i(t + \Delta t) - S_i(t)}{\Delta t} = -\Lambda_i S_i(t) + o(1).$$

Letting $\Delta t \rightarrow 0$ yields

$$S'_i(t) = -\Lambda_i S_i(t), \quad S_i(0) = 1.$$

So

$$S_i(t) = e^{-\Lambda_i t}.$$

The law of the waiting time

From the survival function,

$$\mathbb{P}(\tau_i > t \mid X_0 = i) = e^{-\Lambda_i t}.$$

Hence the cumulative distribution function is

$$\mathbb{P}(\tau_i \leq t \mid X_0 = i) = 1 - e^{-\Lambda_i t}.$$

Differentiating,

$$f_{\tau_i}(t) = \Lambda_i e^{-\Lambda_i t}, \quad t \geq 0.$$

Density

$$f_{\tau_i}(t) = \Lambda_i e^{-\Lambda_i t}.$$

This is exactly the density of an exponential law with parameter Λ_i .

Equivalent small-window form

$$\mathbb{P}(t \leq \tau_i \leq t + \Delta t) \approx e^{-\Lambda_i t} \Lambda_i \Delta t.$$

Survival up to t times jump in the next tiny interval.

CDF, density, and the correct small-window reading

The key small-window interpretation is

$$\mathbb{P}(t \leq \tau_i < t + \Delta t) \approx f_{\tau_i}(t) \Delta t.$$

So once we know the density, we know the probability of the first jump landing near time t .

Since

$$\mathbb{P}(\tau_i > t) = e^{-\Lambda_i t},$$

the CDF is

$$F_{\tau_i}(t) = \mathbb{P}(\tau_i \leq t) = 1 - e^{-\Lambda_i t}, \quad t \geq 0,$$

and the density is

$$f_{\tau_i}(t) = \Lambda_i e^{-\Lambda_i t}.$$

Important clarification

For a continuous random variable,

$$\mathbb{P}(\tau_i = t) = 0.$$

So $f_{\tau_i}(t)$ is not the probability of jumping *exactly* at time t .

Correct interpretation

For small Δt ,

$$\mathbb{P}(t \leq \tau_i < t + \Delta t) \approx f_{\tau_i}(t) \Delta t.$$

It describes the probability of the first jump landing in a tiny interval around t .

A useful factorization of the density

The formula

$$\mathbb{P}(t \leq \tau_i < t + \Delta t) \approx e^{-\Lambda_i t} \Lambda_i \Delta t$$

has a very clean interpretation:

$$\underbrace{e^{-\Lambda_i t}}_{\text{survive until } t \text{ (no jump yet)}} \times \underbrace{\Lambda_i \Delta t}_{\text{jump during the next tiny interval}} .$$

Therefore

$$f_{\tau_i}(t) = S_i(t) \cdot \Lambda_i.$$

What changes between the two probabilities?

- $\Lambda_i \Delta t$: probability of a jump in the next tiny interval, given that we are still in state i now.
- $\Lambda_i e^{-\Lambda_i t} \Delta t$: probability that the *first* jump occurs near time t .

Mean waiting time

Since

$$\tau_i \sim \text{Exp}(\Lambda_i),$$

its expectation is

$$\mathbb{E}[\tau_i] = \int_0^\infty t \Lambda_i e^{-\Lambda_i t} dt = \frac{1}{\Lambda_i}.$$

Interpretation

The total leaving rate Λ_i determines how long the chain typically stays in state i :

large $\Lambda_i \implies$ short waiting time, small $\Lambda_i \implies$ long waiting time.

Equivalent interpretations of the exponential law

Time scale

$$\mathbb{E}[\tau_i] = \frac{1}{\Lambda_i}, \quad \text{Var}(\tau_i) = \frac{1}{\Lambda_i^2}.$$

The parameter Λ_i sets the natural residence-time scale of state i .

Hazard interpretation

$$\lambda_{\text{haz}}(t) = \frac{f_{\tau_i}(t)}{S_i(t)} = \Lambda_i.$$

The instantaneous leaving intensity is constant as long as the chain is still at i .

Memoryless property

$$\mathbb{P}(\tau_i > s + t \mid \tau_i > s) = \mathbb{P}(\tau_i > t).$$

After already waiting for time s , the remaining waiting time has the same law as before.

Conceptual meaning

The state i does not remember how long it has already been occupied; only the current state matters.

“When to jump” and “where to jump” are different objects

The outgoing rates from state i contain two different pieces of information.

1. When to jump

$$\tau_i \sim \text{Exp}(\Lambda_i), \quad \Lambda_i = \sum_{j \neq i} q_{ij}.$$

Only the **sum** of outgoing rates matters for the holding time.

2. Where to jump

Conditioned on the event that a jump occurs, the destination is

$$\mathbb{P}(X_{\tau_i} = j \mid X_0 = i, \text{ jump occurs}) = \frac{q_{ij}}{\Lambda_i}, \quad j \neq i.$$

Only the **relative proportions** of the off-diagonal rates matter.

Do not confuse $\mathbb{P}(\tau_i > t)$ with $P_{ii}(t)$

These two quantities are related but generally not equal:

$$\mathbb{P}(\tau_i > t \mid X_0 = i) = e^{-\Lambda_i t}, \quad P_{ii}(t) = \mathbb{P}(X_t = i \mid X_0 = i).$$

First quantity

The chain has **never left** state i during the whole interval $[0, t]$.

Therefore, in general,

$$\mathbb{P}(\tau_i > t \mid X_0 = i) \leq P_{ii}(t).$$

Second quantity

At time t , the chain is in state i . It may have left earlier and then returned.

First-jump decomposition

Conditioning on the first jump gives a bridge from the waiting-time law to the full finite-time transition matrix.

For $j \neq i$,

$$P_{ij}(t) = \sum_{k \neq i} \int_0^t e^{-\Lambda_i s} q_{ik} P_{kj}(t-s) \, ds.$$

For $j = i$,

$$P_{ii}(t) = e^{-\Lambda_i t} + \sum_{k \neq i} \int_0^t e^{-\Lambda_i s} q_{ik} P_{ki}(t-s) \, ds.$$

How to read the integrand

- $e^{-\Lambda_i s}$: no jump before time s
- $q_{ik} \, ds$: first jump occurs near time s and lands in k
- $P_{kj}(t-s)$: from k , evolve for the remaining time

Summary I

Transition matrix

Starting from the local rule

$$\mathbb{P}(X_{t+\Delta t} = j \mid X_t = i) = q_{ij}\Delta t + o(\Delta t) \quad (j \neq i),$$

we get

$$P(\Delta t) = I + Q\Delta t + o(\Delta t), \quad P'(t) = P(t)Q, \quad P(t) = e^{tQ}.$$

Generator intuition

The off-diagonal entries of Q say where the chain can jump. The diagonal entry says how quickly the current state is abandoned.

Summary II

Waiting time

If the chain starts from i , then the first jump time

$$\tau_i = \inf\{t > 0 : X_t \neq i\}$$

has survival law

$$\mathbb{P}(\tau_i > t \mid X_0 = i) = e^{-\Lambda_i t}, \quad \Lambda_i = \sum_{j \neq i} q_{ij},$$

so

$$\tau_i \sim \text{Exp}(\Lambda_i), \quad \mathbb{E}[\tau_i] = \frac{1}{\Lambda_i}.$$

Final picture

short-time law $\implies$ generator $Q \implies P(t) = e^{tQ} \implies \tau_i \sim \text{Exp}(\Lambda_i)$.